1) Utilice la interacion simple de punto fijo para localizar la raiz de $f(x) = 2\operatorname{sen}\sqrt{x} - x$ haga una eleccion inicial de $x_0 = 0.5$ e itere hasta que error $\leq 0.001\%$. Realizar calculos manuales y en python.

$$2\operatorname{sen}\sqrt{x} - x = 0$$
$$x = 2\operatorname{sen}\sqrt{x}$$

$$E_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100$$

$$x = 2\operatorname{sen}\sqrt{0.5}$$
$$x = 1.299274$$

$$E_a = \left| \frac{1.299274 - 0.5}{1.299274} \right| \times 100$$

$$= 61.51697\%$$

Interacion 1

$$x_1 = 2\operatorname{sen}(\sqrt{0.5}) = 1.299274$$

$$E_a = \left| \frac{1.299274 - 0.5}{1.299274} \right| \times 100 = 61.5169\%$$

Interacion 2:

$$x_2 = 2\operatorname{sen}(\sqrt{1.299274})$$
$$= 1.817147$$

$$E_a = \left| \frac{1.817147 - 1.299274}{1.817140} \right| \times 100$$

$$= 28.499\%$$

Interacion 3

$$X_3 = 2\,\text{sen}\sqrt{1.817147} = 1.95057$$

$$E_a = \left|\frac{1.95057 - 1.817147}{1.95057}\right| \times 100$$

$$= 6.84\%$$

Interacion 4

$$X = 2\,\text{sen}\sqrt{1.95057} = 1.96974$$

$$E_a = \left|\frac{1.96974 - 1.95057}{1.96974}\right| \times 100$$

$$= 0.9732\%$$

Interaccion 5

$$X = 2\,\text{sen}\sqrt{1.96974} = 1.97207$$

$$E_a = \left|\frac{1.97207 - 1.96974}{1.97207}\right| \times 100$$

$$= 0.1181\%$$

Interación 6

$$X = 2\,\text{sen}\sqrt{1.97207} = 1.97234$$

$$E_a = \left|\frac{1.97234 - 1.97207}{1.97234}\right| \times 100$$

$$= 0.0137\%$$

Interación 7

$x = 2 \operatorname{sen} \sqrt{1.97234}$

$= 1.97237$

$$E_a = \left| \frac{1.97237 - 1.97234}{1.97237} \right| \times 100$$

$E_a = 0.0015\%$

interacion 8

$x = 2 \operatorname{sen} \sqrt{1.97237}$

$= 1.972379$

$$E_a = \left| \frac{1.972379 - 1.97237}{1.972379} \right| \times 100$$

0.00045%

$\boxed{1.972379 \text{ alcanza iteración } 8}$